

ENERGY-INEFFICIENT BUILDINGS

A Smart Adaptive Energy Control Approach

Environmental Problem:

Energy-Inefficient Buildings

Sustainable Development Goal:

SDG 17 – Partnerships for the Goals

Proposed Individual Solution:

Smart Adaptive Energy Control System

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1.Introduction

Energy-Inefficient Buildings

Buildings are an essential part of modern society and are used for education, healthcare, offices, housing, shopping and industrial activities. These buildings require considerable amounts of energy for lighting, air conditioning, ventilation, computers, appliances and other electrical systems. When energy is consumed unnecessarily or inefficiently, the building becomes an **energy-inefficient building**.

Energy inefficiency does not necessarily mean that a building has no modern technology. Even a technologically advanced building can waste energy when lights, fans or air-conditioning systems operate when they are not required. For example, an air conditioner may continue cooling an empty classroom, or artificial lights may remain switched on even when sufficient daylight is available.

This problem has environmental as well as economic consequences. Unnecessary electricity consumption increases energy demand and can contribute to greenhouse-gas emissions depending on how the electricity is generated. It also increases the operating cost of buildings.

Addressing this problem requires cooperation among different stakeholders. This is where **SDG 17 – Partnerships for the Goals** becomes relevant. Technology companies, building owners, government organisations, energy providers, researchers and building users can work together to develop and implement effective energy-efficiency solutions.

This report proposes a **Smart Adaptive Energy Control System** that combines technology with stakeholder partnerships to reduce unnecessary energy consumption in buildings.

2. Description of the Environmental Problem

Understanding Energy Inefficient Buildings

An energy-efficient building provides the required services and comfort while using energy effectively. In contrast, an energy-inefficient building consumes more energy than necessary to provide the same services.

The problem can occur in different areas of building operation. Lighting systems may remain active in unoccupied spaces, cooling systems may operate continuously, and electrical equipment may remain powered even when not being used. Poor insulation and building design can further increase the amount of energy required for cooling or heating.

One important reason for energy wastage is the absence of **demand-based control**. Many buildings depend on fixed schedules or manual operation. A fixed schedule assumes that a particular room will always be occupied at a specific time. However, actual occupancy changes throughout the day.

For example, a college classroom may normally be scheduled from 9 AM to 4 PM. However, the room may be empty during breaks, lunch hours or cancelled classes. If the lights and air conditioning remain ON during these periods, energy is consumed without providing any useful service.

Therefore, energy-efficient building management should consider **actual requirements rather than only predetermined schedules**.

The environmental problem can be summarised as:

**Unnecessary Energy Consumption → Increased Energy Demand
→ Greater Environmental Impact**

3.Causes and Impacts

Causes of Energy Inefficiency

Several factors contribute to energy inefficiency in buildings.

1. Unnecessary Lighting

Lights are sometimes operated even when rooms are empty or sufficient natural light is available. This results in avoidable electricity consumption.

2. Excessive Cooling

Air-conditioning systems are major energy consumers. Operating AC systems in empty rooms or maintaining unnecessarily low temperatures can increase energy consumption.

3. Manual Operation

When building systems depend completely on people, devices may be forgotten or operated inefficiently.

4. Fixed Schedules

Timers and fixed schedules cannot always respond to unexpected changes in occupancy.

5. Poor Building Design

Poor insulation, unsuitable windows and insufficient natural ventilation can increase the energy required to maintain comfortable indoor conditions.

Environmental Impacts

Excessive energy consumption increases demand for electricity. Where electricity generation involves fossil fuels, higher electricity demand can be associated with greater greenhouse-gas emissions.

Energy inefficiency can therefore contribute to:

- Increased energy consumption
- Increased operating costs
- Higher electricity demand
- Greater resource consumption
- Increased associated greenhouse-gas emissions
- Reduced overall environmental performance of buildings

Therefore, reducing unnecessary energy use is an important step towards more sustainable building operation.

4.Existing Solutions and Their Limitations

Analysis of Existing Approaches

Several methods are already used to improve building energy efficiency. These include energy audits, energy-efficient appliances, LED lighting, programmable thermostats, timers and energy-monitoring systems.

Energy audits can identify areas where energy is being wasted. However, an audit is generally performed periodically and does not continuously control the building.

Energy-efficient appliances consume less energy than older equipment, but they can still consume electricity unnecessarily if they remain operating when they are not required.

Timers can automatically switch devices according to predetermined schedules. However, they cannot determine whether a room is actually occupied.

Manual switching provides direct control but depends on human behaviour. Users may forget to switch devices off or may not adjust them according to changing conditions.

The **Green Building Energy Management Portal**, which is the expected website solution, can provide a digital platform for monitoring and managing building energy information. However, an individual solution can go one step further by focusing on **automatic real-time control**.

Identified Gap

The main gap is the lack of a system that continuously observes actual conditions and automatically changes energy consumption according to need.

Therefore, the proposed solution focuses on:

Sense → Analyse → Decide → Automatically Control

5. Proposed Individual Solution

Smart Adaptive Energy Control System

The proposed individual solution is a **Smart Adaptive Energy Control System (SAECS)** for buildings.

The system uses sensors and a smart controller to determine the actual requirements of a room. It then automatically controls energy-consuming equipment such as lights, fans and air-conditioning systems.

The system is designed around three important parameters:

Occupancy

Occupancy sensors determine whether people are present in a particular room or area.

Temperature

Temperature sensors measure indoor conditions and provide information that can be used for appropriate cooling control.

Light Availability

Light sensors determine the availability of natural daylight. When adequate daylight is present, artificial lighting can be reduced.

The collected information is processed by a smart controller. Based on predefined control rules, the controller sends commands to smart switches or control units.

Basic System Structure

Occupancy Sensors

Temperature Sensors

Light Sensors



Smart Controller



Decision Making



Lights / Fans / AC



Reduced Unnecessary Energy Consumption

The main difference between this solution and a simple monitoring portal is that the proposed system focuses on **automatic action** rather than only displaying energy information.

6. Working Principle

How the Proposed System Works

The Smart Adaptive Energy Control System operates continuously and responds to changes in building conditions.

Step 1 – Data Collection

Sensors installed inside the building collect information about occupancy, temperature and light levels.

Step 2 – Data Processing

The sensor readings are sent to the smart controller. The controller analyses the information to determine the current requirements of the room.

Step 3 – Decision Making

The controller determines whether lighting, cooling or ventilation needs to be increased, reduced or switched off.

For example, if an area is unoccupied, the system can identify that lighting and cooling are not required at their normal operating level.

Step 4 – Automatic Control

Smart control units then adjust the connected equipment.

For example:

Empty room → unnecessary lights and cooling are reduced.

Occupied room + sufficient daylight → artificial lighting can be reduced.

Occupied room + high temperature → cooling can be adjusted according to the required conditions.

Step 5 – Continuous Adaptation

The system continuously checks the conditions. If people enter or leave the room, or if environmental conditions change, the system can respond accordingly.

Thus, instead of operating according to a fixed schedule, the building becomes **responsive to actual conditions**.

7.SDG 17 Integration

SDG 17 – Partnerships for the Goals

SDG 17 is focused on **strengthening partnerships for sustainable development**. The energy-efficiency problem in buildings cannot always be solved effectively by one stakeholder working independently.

The proposed Smart Adaptive Energy Control System therefore follows a **partnership-based implementation model**.

Building Owners / Institutions

Schools, colleges, offices and other building owners can provide access to the buildings, identify energy problems and support system installation.

Technology Companies

Technology providers can develop and supply sensors, controllers, smart switches and automation technologies.

Energy Providers

Energy organisations can provide energy-consumption information and technical expertise for evaluating energy performance.

Government and Local Authorities

Government organisations can support energy-efficiency policies, environmental regulations, standards and sustainability programmes.

Universities and Research Institutions

Researchers can test the system, analyse energy data and develop improvements to the technology.

Building Users

Students, employees, residents and other users can provide feedback and participate in responsible energy use.

Partnership Model

Government



Building Institution ↔ Technology Company



Energy Provider ↔ Research Institution



Building Users

Together, these stakeholders can support the development, implementation, monitoring and continuous improvement of the solution.

8.Sustainability and Life-Cycle Consideration

Contribution to Sustainable Development

The proposed solution contributes to sustainability by reducing unnecessary energy consumption during the operation of buildings.

Its strongest connection is with **SDG 17**, because successful implementation depends on collaboration, knowledge sharing, technology transfer and resource mobilisation between different organisations.

A technology company may have the technical expertise but may not own buildings. A building institution may have the infrastructure but may not have the required technical knowledge. An energy provider may have useful energy data, while a research institution may have the expertise required to analyse performance.

Therefore, partnership allows these resources to be combined.

Life-Cycle Perspective

The proposed system should also be considered throughout its life cycle:

Manufacturing → Transportation → Installation → Operation → Maintenance → End-of-Life

Electronic sensors and controllers require materials and energy during manufacturing. However, during the operational stage, the system is intended to reduce unnecessary electricity consumption.

Proper maintenance and responsible disposal or recycling of electronic components are also important to reduce environmental impacts at the end of the system's life.

Therefore, sustainability should not only mean reducing energy during operation. It should also involve responsible management of the technology throughout its life cycle.

9. Feasibility and Expected Benefits

Feasibility Analysis

Technical Feasibility

The proposed system can be developed using existing technologies such as occupancy sensors, temperature sensors, light sensors, microcontrollers and smart control units. These technologies can be integrated into a building automation system.

Economic Feasibility

The system requires an initial investment for sensors, controllers, installation and maintenance. However, reducing unnecessary operation of lighting and cooling equipment can potentially reduce electricity consumption and operating costs over time.

A pilot implementation can be carried out in selected classrooms or offices before expanding to the entire building. This reduces initial investment and allows the system to be evaluated.

Operational Feasibility

The system can operate automatically after installation, reducing the need for continuous manual intervention. Building users can still override the system when necessary, depending on the final design.

Expected Benefits

The solution can provide:

- Reduction in unnecessary energy consumption
- Better utilisation of natural daylight
- Reduced operation of cooling systems in unoccupied spaces
- Improved energy efficiency
- Potential reduction in electricity costs
- Better environmental performance
- Data that can help organisations evaluate energy-saving measures

Most importantly, SDG 17 ensures that the solution can be supported by **cooperation between different stakeholders**, making large-scale implementation more practical.

10. Conclusion and Final Proposal

Conclusion

Energy-inefficient buildings are an important environmental problem because they consume energy unnecessarily through inefficient equipment, manual operation, fixed schedules and poor adaptation to actual building conditions.

The proposed **Smart Adaptive Energy Control System** addresses this problem by combining occupancy, temperature and light sensing with intelligent control. Instead of allowing lighting and cooling

systems to operate continuously or according to fixed schedules, the system responds to the actual requirements of the building.

However, technology alone cannot solve the problem on a large scale. The successful implementation of such a system requires cooperation between building owners, technology companies, energy providers, government organisations, research institutions and building users.

This directly connects the proposed solution with **SDG 17 – Partnerships for the Goals**. Through partnerships, different stakeholders can contribute infrastructure, technical knowledge, energy data, funding, research expertise and user feedback.

Final Proposed Model

Energy-Inefficient Buildings



Identify Energy Wastage



Smart Adaptive Energy Control System



Stakeholder Partnerships under SDG 17



Implementation + Monitoring + Improvement



More Efficient and Sustainable Buildings

Final Statement

“Smart energy management becomes more effective when technology, knowledge, resources and people work together.”